Review plan · for the lecturer

Teaching optimisation at university: unit check

Optimization · University · Unit check

Time
40 min

Materials
Projector or board, and the lecture slides.

Learning goals

- Linear programming
- Integer programming
- Heuristics
- Modelling

Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

Discussion

- Which misconception will you target first?
- Which lab will you add?
- How will you assess modelling?

Exit ticket

- One change to my optimisation course this term, and how I will know it worked.

Try it in your next lecture
Bring your open modelling brief and its rubric to your next department meeting for review.

Quiz sheet · for students

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Name _____ Group _____ Date _____

1 **For a bounded LP, an optimum occurs**

◯ At a vertex ◯ Nowhere ◯ Only in the interior

2 **Evaluate the corner.**

3x + 2y at (3, 1) = _____ .

Word bank: 12 · 9 · 11

3 **A constraint with slack at the optimum has a zero shadow price.**

◯ True ◯ False

4 **A maximisation relaxation gives 14. The integer optimum is**

◯ At least 14 ◯ At most 14 ◯ Exactly 14

5 **Compute acceptance.**

Δ = 2, T = 2: p = e^(_____).

Word bank: 1 · -1 · -2

6 **Maximisation: incumbent 13, node bound 12. Branch and bound should**

◯ Branch further ◯ Round the node ◯ Prune the node

7 **Fair heuristic comparisons use which? Choose all that apply.**

☐ Several seeds ☐ Equal time budgets ☐ One best run

8 **Local search always finds the global optimum.**

◯ True ◯ False

9 **Order the modelling cycle.**

☐ Solve

☐ Understand

☐ Validate

☐ Formulate

10 **Best evidence of good modelling:**

◯ Assumptions, validation and defence ◯ The solver's name ◯ The optimal value alone

Answers · for the lecturer

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1 **At a vertex**

Linear objectives peak at corners.

2 **3x + 2y at (3, 1) = 11.**

9 + 2.

3 **True**

Complementary slackness.

4 **At most 14**

The relaxation bounds from above.

5 **Δ = 2, T = 2: p = e^(−1).**

−Δ/T = −1.

6 **Prune the node**

In a maximisation, the bound 12 caps everything below the node, so nothing there can beat 13.

7 **Several seeds; Equal time budgets**

One run is an anecdote.

8 **False**

It stops at local optima.

9 **Top to bottom: 3, 1, 4, 2**

Then revise if needed.

10 **Assumptions, validation and defence**

Grade the model.